

Strategies to manage the usage of mobile in a digital well-being context: Android case study

Mobile monitoring usage user strategies technical DigitalWellbeing android

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Context

Digital wellbeing has emerged as a growing concern as smartphone usage becomes increasingly pervasive and habitual. Research consistently links excessive or unregulated mobile use to reduced attention, sleep disruption, anxiety and diminished quality of life — yet most people have limited awareness of their own usage patterns and limited tools to act on them meaningfully.

Android, as the world's dominant mobile operating system, offers a rich but fragmented ecosystem of APIs and system-level mechanisms that could, in principle, support monitoring and intervention. However, the landscape is complex: permissions are restrictive, OS behaviour varies across manufacturers and versions, and the boundary between useful monitoring and invasive surveillance is thin — particularly in a wellbeing context where user trust and consent are central.

Problem

This proposal sits at the intersection of mobile systems engineering and human-centred design. It takes a deliberate two-step approach: first, map what is technically possible within the Android ecosystem; then, use that map to design and build at least one proof-of-concept intervention that is not only functional but appropriate — privacy-preserving, consent-driven, and grounded in how digital wellbeing is actually understood and measured.

What technical strategies exist on Android to monitor and influence smartphone usage — and can we build a working proof of concept grounded in digital wellbeing principles?

Objectives

One objective is to survey the Android API landscape, assess feasibility and privacy implications, then design and implement at least one intervention-oriented solution evaluated on both technical and user-facing criteria.

1. Review digital wellbeing frameworks and models alongside the Android API landscape, grounding technical choices in validated constructs of problematic smartphone use and healthy digital habits
2. Survey and classify available Android monitoring mechanisms, assessing data accessibility, permissions, privacy implications and suitability for wellbeing-oriented research
3. Select and implement at least one proof-of-concept solution that monitors usage patterns and applies a lightweight, consent-friendly intervention — chosen for both technical feasibility and alignment with wellbeing intervention logic
4. Evaluate the prototype on technical criteria (accuracy, battery impact, robustness) and user-facing criteria (usability, acceptability), with a baseline comparison against built-in Digital Wellbeing

Possible Outcomes (*)

- A structured map of available Android monitoring strategies, with a clear comparison of trade-offs (data richness vs. permissions vs. user friction)
- A working Android prototype demonstrating at least one monitoring or intervention approach

- A validation report comparing the prototype against baseline or gold standard (e.g. built-in Digital Wellbeing, manual logs)
- Documented engineering guidelines for future work — what works, what is blocked by the OS, and where the open problems lie
- A critical review of digital wellbeing frameworks and how they translate — or fail to translate — into technical requirements on a mobile OS

Suggested Work Plan (~9 months)

Period	Phase	Description
Months 1–2	Literature review & scoping	Two parallel tracks: (a) Android API landscape and system architecture; (b) digital wellbeing frameworks, intervention models and problematic smartphone use research. Define research questions, evaluation criteria and ethical considerations.
Month 3	Feasibility assessment	Systematically survey and test available Android APIs. Produce a structured comparison matrix flagging technical trade-offs, permission requirements, OS constraints and wellbeing suitability.
Month 4	Design	Select the most feasible and wellbeing-aligned strategy. Define prototype architecture — monitoring layer, intervention component and privacy model. Validate with supervisor before proceeding.
Months 5–7	Implementation	Build the Android prototype iteratively, stabilising the monitoring layer first then the intervention component. Test on real devices across at least two Android versions. Maintain a development log.
Month 8	Evaluation	Assess technical performance (accuracy, battery, robustness) and user-facing dimensions (usability, acceptability). Compare against built-in Digital Wellbeing as baseline.
Month 9	Writing & consolidation	Interpret findings, reflect on limitations and document engineering and design guidelines. Write and submit the final dissertation.